

Georgia Institute of Technology



Modeling and Optimization of Air Breathing Engines

Final Project Report

AE 4451 - Dr. Seitzman

Executive Overview. This report presents a MATLAB-based cycle analysis and optimization tool for air breathing engine configurations across maximum takeoff thrust, beginning transonic cruise, end transonic cruise, and supersonic cruise. Ramjet, turbojet, and turbofan configurations are modeled using component-level pressure, temperature, work, fuel, and performance relations. The final design study compares condition-specific optimized engines and selects a turbojet-style configuration as the most practical overall design for a supersonic business jet mission profile.

Authors: Setso Kealeboga, Brent Perdew, William Mulder

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Instructor: Dr. Seitzman

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Executive Summary

Our overall optimized engine design maximizes engine performance at any given condition. The conditions being tested in this report are maximum thrust at takeoff (MTO), beginning transonic cruise (BTR), end transonic cruise (ETR), and supersonic cruise (SCR). The corresponding Mach numbers are 0.20, 0.95, 0.95, and 1.6, respectively. The program can determine whether a ramjet, turbojet, or turbofan is optimal for each stage. It breaks down the exact design parameters for each engine design, including bypass ratio, fan pressure ratio, compressor pressure ratio, bleed ratio of air from the compressor into the turbine, combustor fuel ratio, afterburner fuel ratio, and the split of bypass air through each nozzle. From there, it outputs stagnation pressures and temperatures at component exits, nozzle exit velocities, specific thrust, thrust specific fuel consumption, and efficiencies.

The journey from takeoff to supersonic cruise speed has different optimal engine designs depending on the current flight condition. The recommended engine type for the first two conditions is a turbofan, while a turbojet is recommended for the last two. A ramjet is not advised, which makes sense because the aircraft does not approach the Mach number range where a ramjet would become ideal. Even during subsonic flight, the turbofan bypass ratio is relatively low, indicating that bypass flow does not contribute strongly to meeting the specific thrust requirements. All designs are within a reasonable margin of the required specific thrust. The takeoff configuration has a bypass ratio of 1.33, while the beginning transonic cruise configuration has a bypass ratio of 0.15. By the end of transonic cruise, the design has no fan at all.

Despite the different configurations for each flight condition, there is a general increasing trend in overall engine efficiency from approximately 6% to 26% across the four flight conditions.

A turbojet engine is the most versatile option and the best overall design for a business jet. The objective of the aircraft is to accelerate to supersonic speed, maintain that velocity for most of the trip, and then decelerate, descend, and land. The majority of the flight is spent in transonic and supersonic conditions, where a turbojet excels. The tradeoff in takeoff fuel efficiency is offset by the increased efficiency during cruise, ultimately saving fuel compared with a turbofan configuration over the full mission.

Introduction

The purpose of this project is to design and create a computer model to simulate various configurations of air breathing aircraft engines, allowing for rapid engine cycle analysis with varying parameters. This model is then used to identify ideal configurations for given flight conditions and operational requirements.

Engine cycle equations are first derived by breaking the engine into its basic components. Each component has equations for temperature and pressure changes, and these equations are assembled into a single engine function with outputs highlighting the efficiency and performance characteristics of each configuration. Once the equations are derived, they are implemented in MATLAB. Cycle analysis is then performed to provide basic designs for four operating conditions, as well as a fifth overall optimized engine configuration. This process was broken into three main objectives.

The first objective is to find four configurations that match the required specific thrust for the given flight conditions listed in Table 1.1. To accomplish this, the model must quickly iterate through configurations based on the provided design parameter ranges and input conditions until the calculated specific thrust matches the required value.

The second objective is to determine the maximum thrust under the same flight conditions for each configuration found from the first objective, changing only the fuel-to-air and bleed-air ratios. Once a configuration is found that matches the required specific thrust for each flight condition, that configuration is optimized for the maximum possible thrust achievable by that engine. Specific fuel consumption at this maximum thrust is also calculated and checked for practical reasonableness. The compressor pressure ratio, fan pressure ratio, bypass ratio, and nozzle split ratio remain fixed during this step.

The third objective is to optimize a single configuration that best fits all four conditions and requirements. In reality, a single engine configuration is used over the full flight. Configurations are therefore tested to meet the requirements from the first objective while also providing an ideal configuration that is operationally effective and practical across the full mission.

Table 1.1: Requirements and Flight Conditions for Objective 1

Flight Condition	Altitude	T_a	P_a	M	Required Specific Thrust kN-s/kg lb _F S/lb _m
	km/kft	K/R	kPa/psia		
Max Takeoff Thrust (MTO)	0	288/518	101.3/14.7	0.20	1.23/125
Begin Transonic Cruise (BTR)	10.0/32.8	223/402	26.4/3.83	0.95	1.05/107
End Transonic Cruise (ETR)	10.0/32.8	223/402	26.4/3.83	0.95	1.46/149
Supersonic Cruise (SCR)	16.5/53.5	216/389	9.81/1.42	1.6	0.874/89.1

The inputs to the model, in addition to the flight conditions, are design parameters that can be altered to meet the specific thrust requirements. These include the stagnation pressure ratios of the compressor and fan, the fuel-air ratios for the main burner and afterburner, the ratios of bypass and bleed air, and the split ratio indicating whether a split nozzle is used. Component characteristics and properties used in the governing equations include average normalized specific heat, adiabatic efficiency, polytropic efficiency, combustion efficiency, and fuel characteristics. A complete list is included in Appendix C. Constraints are also placed on maximum burner and afterburner temperatures, maximum bleed air and bypass ratios, minimum and maximum fan pressure ratios, and maximum overall compression ratio.

From these inputs, the model can simulate a ramjet, turbojet, or turbofan engine. It outputs airspeed, exit stagnation temperature and pressure for all components except the nozzles and fuel pump, exhaust temperature, exhaust pressure, exhaust Mach number, power per mass core air flow rate, specific thrust, specific fuel consumption, maximum burner and afterburner temperatures, fuel-air ratios, and propulsive, thermal, and overall efficiency.

Approach

The first step was to develop the governing equations. Every engine component changes at least the pressure and temperature of the mass flowing through it. These equations act as building blocks for the full engine model. By programming these blocks into MATLAB as individual functions, they can be assembled in the same order that air flows through an engine.

The simulation is structured so that every component produces an output, regardless of whether that component is active for a given configuration. This behavior is controlled through design inputs such as bypass air ratio and fuel ratios. For example, if the bypass air ratio through the fan is zero, the fan is effectively absent and the engine becomes either a ramjet or turbojet. Similarly, the parameter σ , ranging from 0 to 1, determines whether a split nozzle or combined nozzle is used. This allows a single station numbering system and a single engine function to represent the full range of configurations.

A component diagram was created to show how each component interacts with the others and to verify that each component receives the correct inputs and sends outputs to the correct downstream components. This diagram is included in Appendix A.

Using conservation principles, the pressure and temperature changes through each component were calculated. Conservation of momentum was used for pressure changes. For components that require or output work, conservation of energy was used to calculate work and temperature changes. Energy conservation was also used to calculate stagnation temperature changes for all components. Outputs from one component become inputs to the next, allowing a unified engine function to calculate the complete set of output values.

Several assumptions were used to simplify the model. All components were assumed to be adiabatic, so no heat transfer occurs between the engine and surroundings. The nozzles were assumed to be perfectly expanded, even at supersonic speeds, meaning the nozzle exit pressure equals ambient pressure. The adiabatic efficiencies for the fan, compressor, and turbines were estimated using polytropic efficiency, as shown in Appendix D. Shaft and transmission efficiency for the turbines were assumed to be 100%, meaning there was no power transmission loss between turbines and the components they drive.

Additional assumptions were made for pressure and temperature behavior. Fuel was assumed to be stored at a pressure higher than ambient, so the pump must produce a pressure greater than the main combustor inlet pressure. For supersonic flight conditions, diffuser exit stagnation pressure was multiplied by a ram recovery factor. For turbofan configurations, an estimated drag factor was added based on engine size, which was related

to bypass ratio. Maximum temperature was estimated based on the ratio of compressor bleed air to maximum bleed air. Finally, the exit stagnation pressure of a combined nozzle was calculated by multiplying the reversible combined nozzle stagnation pressure by a pressure ratio representing fractional stagnation pressure loss.

After the equations were developed, the team integrated the component codes into a single engine function. This was challenging because the code was developed by three different people with different coding styles and output preferences, including vectors versus structures. Eventually, the individual component codes were combined into a completed MATLAB function that could be used to solve the three project objectives.

The four flight-condition configurations were identified using MATLAB's `fmincon()` function. Upper and lower bounds were defined for each design parameter, and the required specific thrust was imposed as a constraint. Additional constraints from Appendix C were also included. Since `fmincon()` is limited to local minima, the resulting designs are feasible but not necessarily globally optimal.

Once the four configurations were found, the parameters were entered into the engine model to compute component temperatures and pressures, output velocities, powers, maximum temperatures, and fuel-air ratios. These configurations were then optimized to find the minimum negative specific thrust, which corresponds to maximum specific thrust. The associated SFC values were also calculated. Finally, a single configuration was selected as the best overall fit for all flight conditions based on the mission intent and then optimized by changing selected design parameters.

Results

3.1 Configuration 1: Engine for MTO

The first configuration corresponds to maximum takeoff thrust at Mach 0.20. The optimized design parameters are shown in Table 3.1. The configuration uses a turbofan layout with nonzero bypass flow and no afterburner.

Table 3.1: Design Parameters for Configuration 1

β	Pr_f	Pr_c	b	f	f_{ab}	σ
1.33312	1.3842	44.8467	0.1499	0.02720	0	0.500

Table 3.2: Output Velocity, Power, and Fuel Ratios for Configuration 1

u_e	u_{ec}	u_{ef}	M_e	M_{ec}	M_{ef}	f_{max}	$f_{ab,max}$	Turbine Power (kJ/kg)
(m/s)	(m/s)	(m/s)						
1124.3	783.3	243.7	2.02	1.51	0.71	0.0272	0.0312	725.9

Fan Turbine Power (kJ/kg)
3075.0

Table 3.3: Performance Parameters for Configuration 1

ST (kN-s/kg)	TSFC (kg/kN-hr)	η_{th}	η_p	η_o
1.23	45.98	54.64%	11.41%	6.23%

Table 3.4: Output Stagnation Temperature and Pressure for Configuration 1

Station	1	2	3	4	5.1	5.1m	5.2	6	7	e^1	ef^1	ec^1	f.1	f.2
T_o (K)	290.3	320.8	1017.9	2099.2	1475.8	1410.7	1351.5	1351.5	971.4	813.6	291.4	783.3	-	-
p_o (kPa)	104.0	143.5	643.6	611.5	108.4	112.1	93.0	90.2	393.93	101.3	101.3	101.3	122	7008.3

¹ Nozzle exhaust temperature and pressure values are static, not stagnation.

Table 3.5: Max Thrust for Configuration 1 at Each Flight Condition

Flight Condition	MTO	BTR	ETR	SCR
Max possible ST	1.08	0.968	0.968	0.869
SFC at Max ST	90.4	101.2	101.2	112.7

3.2 Configuration 2: Engine for BTR

The second configuration corresponds to the beginning of transonic cruise at Mach 0.95. The design keeps a small amount of bypass flow and uses an afterburner fuel ratio of 0.0137.

Table 3.6: Design Parameters for Configuration 2

β	Pr_f	Pr_c	b	f	f_{ab}	σ
0.15	1	40.91	0.15	0.0312	0.0137	0.451

Table 3.7: Output Velocity, Power, and Fuel Ratios for Configuration 2

u_e	u_{ec}	u_{ef}	M_e	M_{ec}	M_{ef}	f_{max}	$f_{ab,max}$	Turbine Power (kJ/kg)
(m/s)	(m/s)	(m/s)						
1130.2	535.74	339.48	1.999	1.448	1.120	0.0294	0.0305	376.13

Fan Turbine Power (kJ/kg)
22.43

Table 3.8: Performance Parameters for Configuration 2

ST (kN-s/kg)	TSFC (kg/kN-hr)	η_{th}	η_p	η_o
1.05	137.91	42.98%	39.73%	17.08%

Table 3.9: Output Stagnation Temperature and Pressure for Configuration 2

Station	1	2	3	4	5.1	5.1m	5.2	6	7	e^1	ef^1	ec^1	f.1	f.2
T_o (K)	263.25	285.53	646.71	2041.7	1722.4	1587.3	1381.2	1381.2	482.44	840.27	840.27	342.37	-	-
p_o (kPa)	46.59	59.35	877.10	833.25	362.77	435.97	233.95	226.93	95.19	26.4	26.4	26.4	47.1	1449.1

¹ Nozzle exhaust temperature and pressure values are static, not stagnation.

Table 3.10: Max Thrust for Configuration 2 at Each Flight Condition

Flight Condition	MTO	BTR	ETR	SCR
Max possible ST	1.418	1.315	1.315	1.215
SFC at Max ST	114	123	123	133

3.3 Configuration 3: Engine for ETR

The third configuration corresponds to the end of transonic cruise. The optimized design no longer uses a fan, so the bypass ratio is zero and the nozzle split parameter is not applicable.

Table 3.11: Design Parameters for Configuration 3

β	Pr_f	Pr_c	b	f	f_{ab}	σ
0	1	28.83	0.15	0.03368	0.01511	N/A

Table 3.12: Output Velocity, Power, and Fuel Ratios for Configuration 3

u_e	u_{ec}	u_{ef}	M_e	M_{ec}	M_{ef}	f_{max}	$f_{ab,max}$	Turbine Power (kJ/kg)
(m/s)	(m/s)	(m/s)						
362.1	591.3	298.3	1.94	3.23	0.987	0.0298	0.287	588.9

Fan Turbine Power (kJ/kg)
7963

Table 3.13: Performance Parameters for Configuration 3

ST (kN-s/kg)	TSFC (kg/kN-hr)	η_{th}	η_p	η_o
1.46	342.3	47.28%	14.55%	6.88%

Table 3.14: Output Stagnation Temperature and Pressure for Configuration 3

Station	1	2	3	4	5.1	5.1m	5.2	6	7	e^1	ef^1	ec^1	f.1	f.2
T_o (K)	263.25	271.17	836.6	1410	847	845.4	837	1210.1	1201.5	1222.6	298.3	591.3	-	-
p_o (kPa)	45.69	50.26	2056	1953	199.3	199.3	191.5	185.8	125.5	26.4	26.4	26.4	47.1	2628.1

¹ Nozzle exhaust temperature and pressure values are static, not stagnation.

Table 3.15: Max Thrust for Configuration 3 at Each Flight Condition

Flight Condition	MTO	BTR	ETR	SCR
Max possible ST	4.21	4.89	4.89	5.23
SFC at Max ST	397.1	342.32	342.32	319.9

3.4 Configuration 4: Engine for SCR

The fourth configuration corresponds to supersonic cruise at Mach 1.6. The design is a turbojet-style configuration with no bypass flow and no afterburner.

Table 3.16: Design Parameters for Configuration 4

β	Pr_f	Pr_c	b	f	f_{ab}	σ
0	1	27.69	0.15	0.0265	0	N/A

Table 3.17: Output Velocity, Power, and Fuel Ratios for Configuration 4

u_e	u_{ec}	u_{ef}	M_e	M_{ec}	M_{ef}	f_{max}	$f_{ab,max}$	Turbine Power (kJ/kg)
(m/s)	(m/s)	(m/s)						
5503.0	5503.0	-	3.25	3.25	-	0.0405	6.93	698.04
			Fan Turbine Power (kJ/kg)					
			589.11					

Table 3.18: Performance Parameters for Configuration 4

ST (kN-s/kg)	TSFC (kg/kN-hr)	η_{th}	η_p	η_o
0.874	149.9	65.13%	0.04%	26.04%

Table 3.19: Output Stagnation Temperature and Pressure for Configuration 4

Station	1	2	3	4	5.1	5.1m	5.2	6	7	e^1	ef^1	ec^1	f.1	f.2
T_o (K)	326.59	-	1055.4	927.9	930.2	933.3	933.5	933.5	933.5	1008	-	1008	-	-
p_o (kPa)	37.35	-	1758.0	1670.1	1301.7	290.6	285.3	276.7	276.7	9.81	-	9.81	30.51	2330

¹ Nozzle exhaust temperature and pressure values are static, not stagnation.

Table 3.20: Max Thrust for Configuration 4 at Each Flight Condition

Flight Condition	MTO	BTR	ETR	SCR
Max possible ST	6.07	6.23	6.23	6.36
SFC at Max ST	157	153	153	150

Discussion and Overall Optimized Engine

At a Mach number of 0.2, corresponding to maximum takeoff thrust, no afterburner is used. This is logical because the afterburner would consume more fuel than the benefit of its added thrust is worth, in addition to the practical noise concerns associated with afterburners. A high propulsive efficiency is also desired at takeoff. An afterburner increases the exit velocity of air through the engine, which lowers propulsive efficiency. Another important design parameter is that half of the bypass air from the fan goes through the core nozzle while the other half leaves through a separate nozzle, meaning some of the air moving through the core can be cooled. The main burner fuel consumption is somewhat high, but that is expected for takeoff, when the aircraft must accelerate from rest.

At Mach 0.95, corresponding to the beginning of transonic cruise, the difference between exit velocity and flight speed is less important to efficiency than it is at takeoff. It is still slightly more efficient to use an engine with a fan, but only with a fraction of the air flow through the core. Otherwise, the engine size would create more drag than the fan thrust is worth. This amount of bypass air is enough to route some flow through the core nozzle, improving propulsive efficiency and slightly lowering thrust specific fuel consumption. Using an afterburner for subsonic flight would be extremely inefficient. More thrust is gained from having a slightly higher fuel ratio than from adding fuel in the afterburner. The core pressure ratio is increased to improve cycle thermal efficiency, and compressor bleed safely raises the turbine inlet temperature.

As transonic flight ends and supersonic flight begins, the optimal engine changes from a turbofan to a turbojet. As velocity increases, the fan contributes significant drag while bypass air contributes little to increasing specific thrust. It is instead more efficient to use afterburning temporarily to help the aircraft cross the sound barrier. Since this is only a temporary operating condition and most of the flight is spent supersonic with the afterburner off, the overall design remains effective even though this configuration has poor fuel efficiency. The maximum compressor bleed ratio is recommended to raise the maximum combustor temperature as much as possible.

For supersonic cruise, the optimal bypass ratio is zero. A turbofan does not provide the best fuel efficiency for the desired specific thrust because the additional exit velocity from bypass flow does not significantly improve thermal efficiency. Any increase in engine size would mainly add drag. A higher compressor pressure ratio is logical because it increases

cycle thermal efficiency. As a result, the fuel efficiency at cruise speed is better than at the BTR flight condition, since the same amount of thrust can be produced with less fuel input. The maximum amount of air is bled from the compressor into the turbine to raise the maximum temperature leaving the combustor. Since the vehicle is in cruise, afterburner use is unnecessary.

4.1 Overall Optimized Engine

The engine with performance parameters similar to the third and fourth flight conditions was selected as the overall optimal engine. These flight conditions represent a business jet mission profile in which the aircraft takes off, accelerates quickly to supersonic speed, and then cruises for most of the trip. Any flight worth taking with a supersonic aircraft is likely long enough that most of the mission is spent cruising rather than taking off or descending. An engine without a fan produces the highest efficiencies during transonic and supersonic flight because minimizing drag is more important than the small benefit from bypass nozzle exit velocity. Even for the beginning transonic cruise configuration, the bypass ratio is small, indicating limited benefit from the fan. Therefore, the turbojet configuration has the greatest fuel efficiency for the majority of the mission.

Table 4.1: Optimized Engine Design Fixed Parameters

β	Pr_f	Pr_c	σ
0	1	28.83	N/A

Table 4.2: Performance Parameters for Optimized Engine at Each Flight Condition

	MTO	BTR	ETR	SCR
b	0.05	0.1	0.15	0.15
f	0.03	0.025	0.0308	0.012
f_{ab}	0	0.012	0.01511	0
ST	1.1	1.4	5.1	6.4
TSFC	100	133	363	154

Engine Model Diagram

Appendix A – Engine model diagram

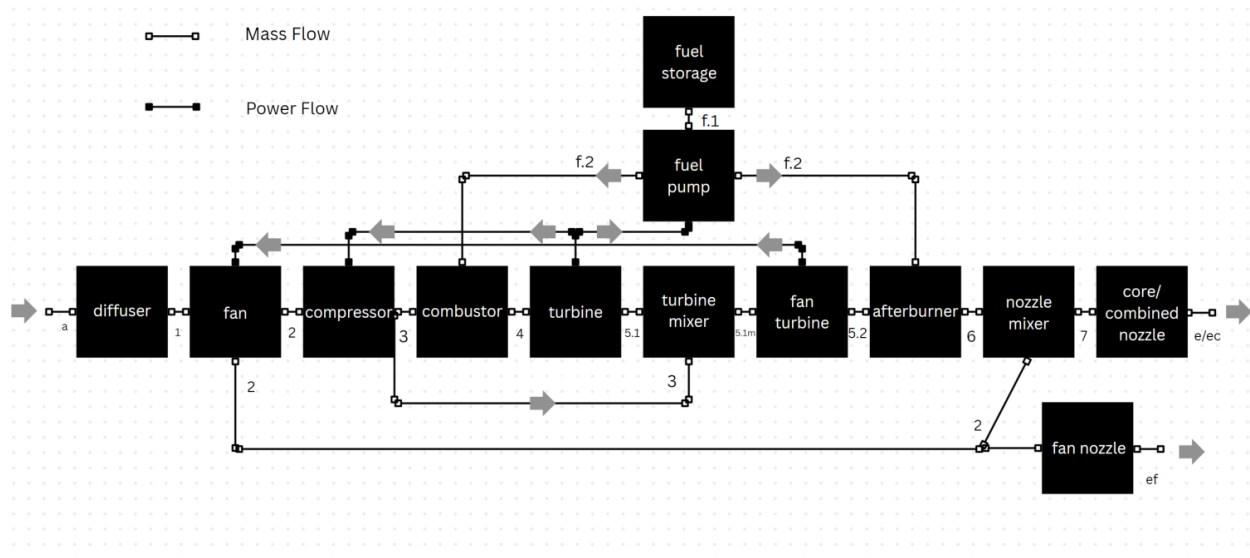


Figure A.1: Engine model diagram showing mass flow, power flow, fuel storage, fuel pump, diffuser, fan, compressor, combustor, turbine stages, afterburner, nozzle mixer, combined nozzle, and fan nozzle.

Equations

Fuel Pump

$$p_{of.2} = p_{injection} + p_{o3} \quad (B.1)$$

$$\dot{w}_{fp} = \left(\frac{p_{of.2} - p_{of.1}}{\rho_f \eta_{pump}} \right) (f + f_{ab}) \quad (B.2)$$

Diffuser

$$p_{o1} = p_a r_d \left[1 + \eta_d \left(\frac{\gamma - 1}{2} \right) M^2 \right]^{\gamma/(\gamma-1)} \quad (B.3)$$

$$T_{o1} = T_{oa} \quad (B.4)$$

Fan

$$p_{o2} = p_{o1} Pr_f \quad (B.5)$$

$$T_{o2} = T_{o1} \left[1 + \frac{1}{\eta_f} \left(Pr_f^{(\gamma-1)/\gamma} - 1 \right) \right] \quad (B.6)$$

$$\frac{\dot{W}_f}{\dot{m}_a + \dot{m}_s} = C_{p,f} (T_{o2} - T_{o1}) \quad (B.7)$$

Compressor

$$p_{o3} = p_{o2} Pr_c \quad (B.8)$$

$$T_{o3} = T_{o2} \left[1 + \frac{1}{\eta_c} \left(Pr_c^{(\gamma-1)/\gamma} - 1 \right) \right] \quad (B.9)$$

$$\frac{\dot{W}_c}{\dot{m}_a} = C_{p,c} (T_{o3} - T_{o2}) (1 + \beta) \quad (B.10)$$

Combustor

$$p_{o4} = Pr_b p_{o3} \quad (\text{B.11})$$

$$T_{o4} = \left(\frac{1}{1 + f - b} \right) \left(\frac{\eta_b f \Delta h_R}{C_{p,b}} \right) + T_{o3}(1 - b) \quad (\text{B.12})$$

$$T_{max} = T_{max,0} + C_b \left(\frac{b}{b_{max}} \right)^n \quad (\text{B.13})$$

$$f_{max} = [(T_{max}(1 + f - b)) - T_{o3}(1 - b)] \left(\frac{C_{p,b}}{\Delta h_R \eta_b} \right) \quad (\text{B.14})$$

Turbine

$$p_{o5.1} = p_{o4} \left(\frac{\frac{T_{o5.1} - T_{o4}}{\eta_t} + T_{o4}}{T_{o4}} \right)^{C_{p,t}/R} \quad (\text{B.15})$$

$$T_{o5.1} = T_{o4} - \frac{\dot{W}_c + \dot{W}_{fp}}{(1 + f - b)C_{p,t}} \quad (\text{B.16})$$

$$\dot{W}_t = \dot{W}_c + \dot{W}_{fp} \quad (\text{B.17})$$

Mixer Turbine

$$p_{o5.1m} = p_{o5.1} \left(\frac{T_{o5.1m}}{T_{o5.1}} \right)^{(C_{p,hot}/R)(1+f-b)} \left(\frac{T_{o5.1m}}{T_{o3}} \right)^{(C_{p,cold}/R)b} \quad (\text{B.18})$$

$$T_{o5.1m} = \frac{C_{p,cold}(bT_{o3}) + C_{p,hot}((1 + f - b)T_{o5.1})}{C_{p,hot}(1 + f - b) + C_{p,cold}b} \quad (\text{B.19})$$

Fan Turbine

$$p_{o5.2} = p_{o5.1} \left[1 - \frac{1}{\eta_{p,ft}}(1 - T_r) \right]^{C_{p,ft}/R} \quad (\text{B.20})$$

$$T_{o52} = T_{o51} - \frac{\dot{W}_{ft}(1 - \beta)}{C_{p,ft}/R} \quad (\text{B.21})$$

Afterburner

$$T_{o6} = \frac{(1+f)C_p T_{o5.2} + \eta_{ab} f_{ab} h_R}{(1+f+f_{ab})C_p} \quad (\text{B.22})$$

$$P_{o6} = P_{o5.2} P_{r_{ab}} \quad (\text{B.23})$$

$$f_{max,ab} = \frac{(1+f)C_{p,ab} T_{max,ab} - T_{o6}}{\eta_{ab} h_R - C_{p,T_{max}} T_{max,ab}} \quad (\text{B.24})$$

Combined Nozzle

$$T_{ec,is} = T_{o7} \left(\frac{P_a}{P_{o7}} \right)^{R/C_p} \quad (\text{B.25})$$

$$T_{ec} = T_{o7} - \eta_{cn} (T_{o7} - T_{ec,is}) \quad (\text{B.26})$$

$$u_{ec} = [2C_p(T_{o7} - T_{ec})]^{1/2} \quad (\text{B.27})$$

$$a_{ec} = \left[\frac{C_p/R}{C_p/R - 1} \frac{R}{MW} T_{ec} \right]^{1/2} \quad (\text{B.28})$$

$$M_{ec} = \frac{u_{ec}}{a_{ec}} \quad (\text{B.29})$$

Core Nozzle

$$T_{e,is} = T_{o7} \left(\frac{P_a}{P_{o7}} \right)^{R/C_p} \quad (\text{B.30})$$

$$T_e = T_{o7} - \eta_n (T_{o7} - T_{e,is}) \quad (\text{B.31})$$

$$u_e = [2C_p(T_{o7} - T_e)]^{1/2} \quad (\text{B.32})$$

$$a_e = \left[\frac{C_p/R}{C_p/R - 1} \frac{R}{MW} T_e \right]^{1/2} \quad (\text{B.33})$$

$$M_e = \frac{u_e}{a_e} \quad (\text{B.34})$$

Fan Nozzle

$$T_{ef,is} = T_{o7} \left(\frac{P_a}{P_{o7}} \right)^{R/C_p} \quad (\text{B.35})$$

$$T_{ef} = T_{o7} - \eta_{fn}(T_{o7} - T_{ef,is}) \quad (\text{B.36})$$

$$u_{ef} = [2C_p(T_{o7} - T_{ef})]^{1/2} \quad (\text{B.37})$$

$$a_{ef} = \left[\frac{C_p/R}{C_p/R - 1} \frac{R}{MW} T_{ef} \right]^{1/2} \quad (\text{B.38})$$

$$M_{ef} = \frac{u_{ef}}{a_{ef}} \quad (\text{B.39})$$

Performance Parameters

$$ST = (1 + f + f_{ab})u_e - u - C_\beta M^2 \left(\frac{p_a}{p_{STP}} \right) \beta^{1.5} \quad (\text{B.40})$$

$$TSFC = \frac{f + f_{ab}}{ST} 3600 \quad (\text{B.41})$$

$$\eta_{th} = \frac{(1 + f + f_{ab})u_e^2 - u^2}{2(f + f_{ab})\Delta h_R} \quad (\text{B.42})$$

$$\eta_o = \frac{u}{(TSFC/3600000)\Delta h_R} \quad (\text{B.43})$$

$$\eta_p = \frac{\eta_o}{\eta_{th}} \quad (\text{B.44})$$

Nozzle Mixer

$$\dot{m}_a = 1 + f + f_{ab} \quad (\text{B.45})$$

$$\dot{m}_s = (1 - \sigma)\beta \quad (\text{B.46})$$

$$\dot{m}_7 = \dot{m}_a + \dot{m}_s \quad (\text{B.47})$$

$$T_{o7} = \frac{\dot{m}_a C_{p,a} T_{o6} + \dot{m}_s C_{p,s} T_{o2}}{\dot{m}_a C_{p,a} + \dot{m}_s C_{p,s}} \quad (\text{B.48})$$

$$P_{o7} = Pr_{nm} \left(P_{o2}^{\dot{m}_s/\dot{m}_7} \right) \left(P_{o6}^{\dot{m}_a/\dot{m}_7} \right) \left(\frac{T_{o7}}{T_{o6}} \right)^{(C_{p,a}/R)(\dot{m}_a/\dot{m}_7)} \left(\frac{T_{o7}}{T_{o2}} \right)^{(C_{p,s}/R)(\dot{m}_s/\dot{m}_7)} \quad (\text{B.49})$$

Characteristics of the Turbine Engine Components

Table C.1: Characteristics of the Turbine Engine Components

Component	η_i	Pr_i	MolecWt	c_p/R	Other
Diffuser (d)	0.94*		28.9	3.5	
Fan (f)	0.92**	$1.15 < Pr_f \leq 1.7$ if used	28.9	3.5	$\beta_{max} = 15$, $C_\beta = 0.263$ kN-s/kg; if $\beta = 0$, fan cannot be used.
Compressor (c)	0.91**	$Pr_c < 55/Pr_f$	28.9	3.62	
Main Burner (b)	0.99	0.95 if used	28.9	Note 1	Heating value $f_{uel} = 43.52$ MJ/kg or 18,710 BTU/lbm.
Turbine (t)	0.94**		28.9	Note 2	$T_{max,0} = 1400$ K, $b_{max} = 0.15$, $n = 0.6$, $C_b = 700$ K.
Turbine Mixer (tm)			28.9	Note 3	
Fan Turbine (ft)	0.94**		28.9	Note 4	
Afterburner (ab)	0.96	0.97 if used	28.9	Note 5	Same heating value as main burner; $T_{max,ab} = 2300$ K.
Core Nozzle (n)	0.96*		28.9	Note 6	
Fan/Bypass Nozzle (fn)	0.97* if used		28.9	3.5	
Nozzle Mixer (nm)			28.9	Note 7	$C_{nm} = 2.00$.
Combined Nozzle (cn)	0.96*		28.9	Note 8	
Fuel Pump (p)	0.48*				$\rho_f = 780$ kg/m ³ , $\Delta p_{f,1} = 20.7$ kPa, $\Delta p_{inj} = 572$ kPa.

*Adiabatic/isentropic efficiency. **Polytropic efficiency.

$$^1 c_{p,b}/R = 3.70 + 0.660(T_{o3}/1000K)^2 - 0.200(T_{o3}/1000K)^3.$$

$$^2 c_{p,t}/R = 3.38 + 0.700(T_{o4}/1000K)^2 - 0.200(T_{o4}/1000K)^3.$$

$$^3 \text{ Hot flow from turbine: } c_{p,tm-hot}/R = 3.43 + 0.780(T_{o5.1}/1000K)^2 - 0.270(T_{o5.1}/1000K)^3.$$

$$\text{Cold/bleed flow from compressor: } c_{p,tm-cold}/R = 3.70 + 0.780(T_{o3}/1000K)^2 - 0.360(T_{o3}/1000K)^3.$$

$$^4 c_{p,ft}/R = 3.40 + 0.630(T_{o5.1m}/1000K)^2 - 0.200(T_{o5.1m}/1000K)^3.$$

$$^5 c_{p,ab}/R = 3.50 + 0.720(T_{o5.2}/1000K)^2 - 0.210(T_{o5.2}/1000K)^3.$$

$$^6 c_{p,n}/R = 3.45 + 0.550(T_{o6}/1000K)^2 - 0.150(T_{o6}/1000K)^3.$$

$$^7 \text{ Core flow: } c_{p,nm-core}/R = 3.44 + 0.790(T_{o,core}/1000K)^2 - 0.270(T_{o,core}/1000K)^3. \text{ Bypass flow: } c_{p,nm-by}/R = 3.43 + 0.790(T_{o,by}/1000K)^2 - 0.280(T_{o,by}/1000K)^3, \text{ where } T_{o,core} = (T_{o6} + T'_{o7})/2, T_{o,by} = (T_{o2} + T'_{o7})/2, \text{ and } T'_{o7} \text{ is the estimated nozzle mixer output temperature if both flows had the same } c_p/R.$$

$$^8 c_{p,cn}/R = 3.45 + 0.550(T_{o7}/1000K)^2 - 0.150(T_{o7}/1000K)^3.$$

Assumption Equations

Adiabatic Efficiency of a Compressor or Fan

$$\eta_i = \frac{Pr_i^{R/c_p} - 1}{Pr_i^{(R/c_p)/\eta_{p,i}} - 1} \quad (\text{D.1})$$

Adiabatic Efficiency of a Turbine

$$\eta_i = \frac{Tr_i - 1}{Tr_i^{1/\eta_{p,i}} - 1} \quad (\text{D.2})$$

Ram Recovery Factor

$$p_{o1} = r_d p'_{o1} \quad (\text{D.3})$$

$$r_d = \begin{cases} 1, & M \leq 1 \\ 1 - 0.075(M - 1)^{1.35}, & 1 < M < 5 \end{cases} \quad (\text{D.4})$$

Maximum Turbine Inlet Temperature Based on Bleed Air Ratio

$$T_{max} = T_{max,0} + C_b \left(\frac{b}{b_{max}} \right)^n \quad (\text{D.5})$$

Specific Drag Increase Based on Fan Bypass Ratio

$$\Delta d = C_\beta M^2 \left(\frac{p_a}{p_{STP}} \right) \beta^{1.5} \quad (\text{D.6})$$

Exit Stagnation Pressure of Irreversible Combined Nozzle

$$p_{o7} = p_{o7,rev} Pr_{nm} \quad (D.7)$$

$$Pr_{nm} = e^{-C_{nm}/(1+m_r^{0.5})}, \quad m_r = \frac{\dot{m}_{max}}{\dot{m}_{min}} \quad (D.8)$$